Satellite Accounts

Lecturer: Dr Amanda Irwin
Unit coordinator: Prof. Arunima Malik
amanda.irwin@sydney.edu.au
arunima.malik@sydney.edu.au

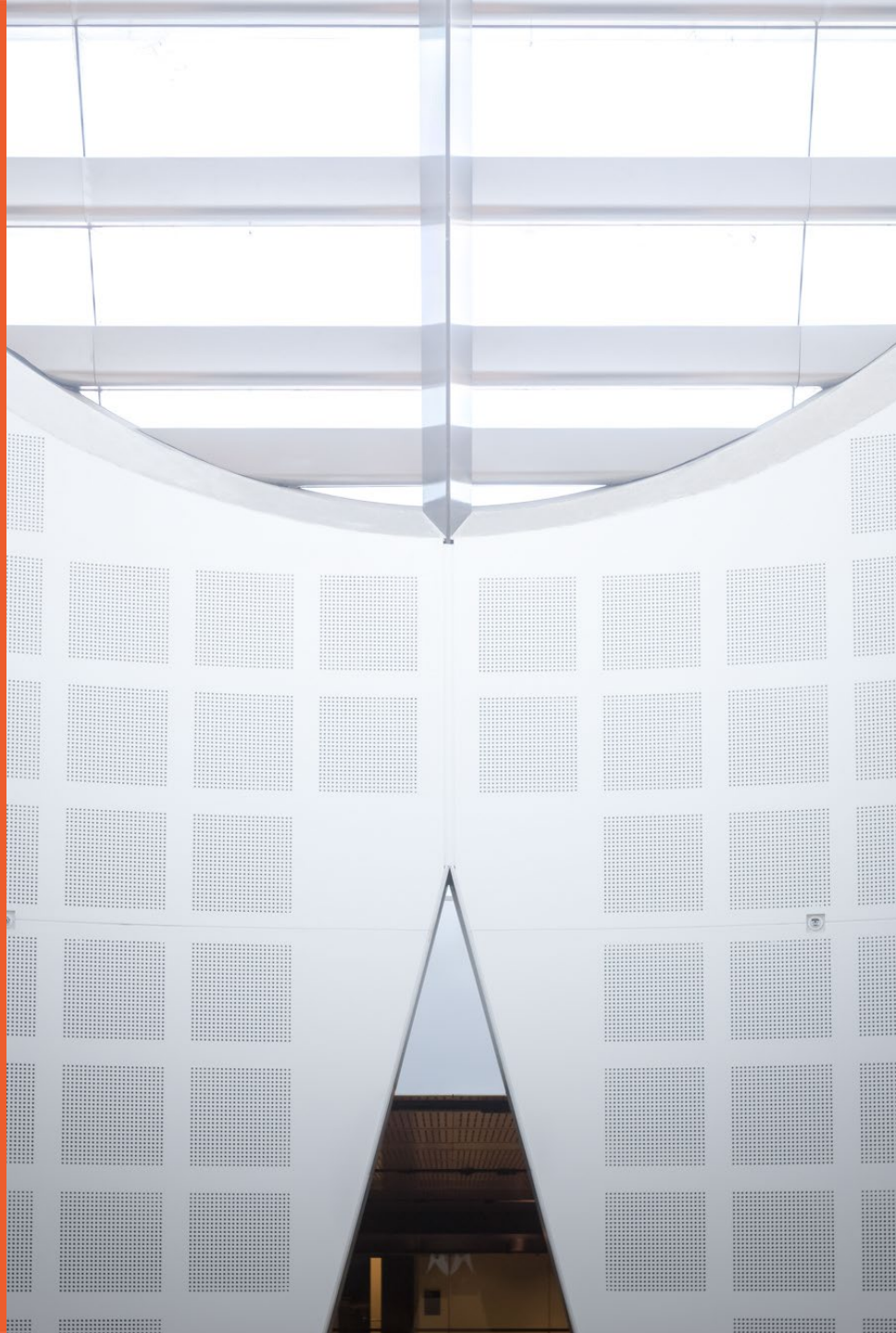
THE UNIVERSITY OF SYDNEY



We recognise and pay respect to the Elders and communities – past, present, and emerging – of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

THE UNIVERSITY OF SYDNEY

PHYS5033



THE UNIVERSITY OF SYDNEY

Recap – Week 4

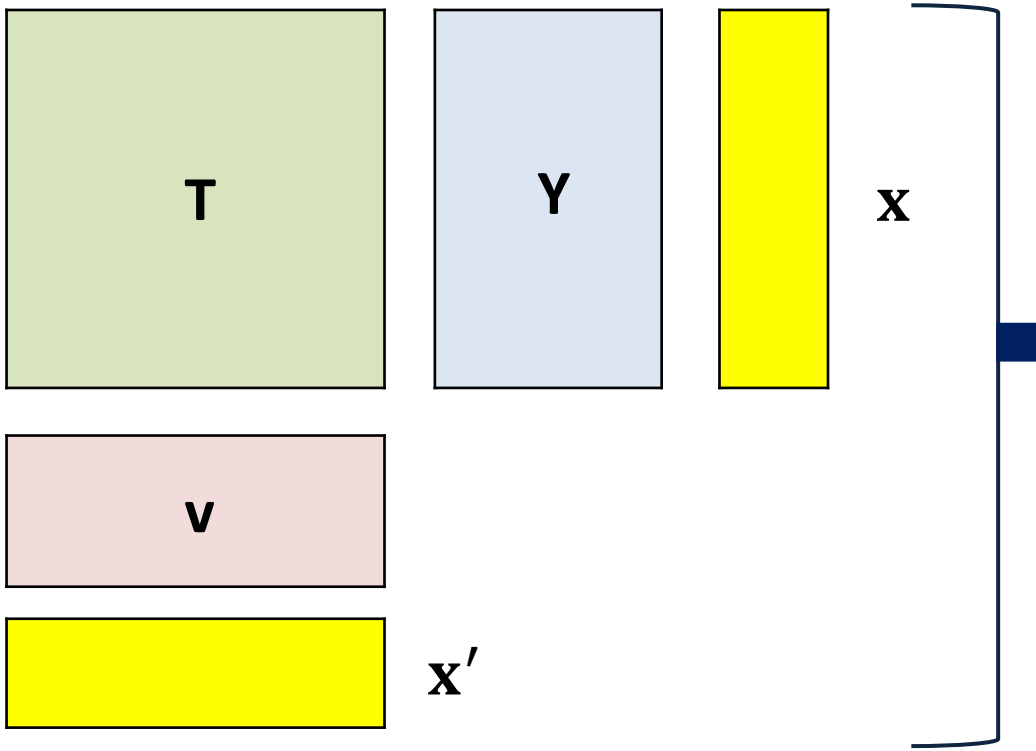
- Matrix diagonalisation
- Matrix inversion
- Direct requirements matrix – direct inputs required to produce $1 of total output
- Total requirements matrix – total inputs required to produce $1 of total output
- Fundamental input-output equation

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{y}$$

Total output | L matrix Leontief inverse | Total final demand

$$\mathbf{x} = \mathbf{L}\mathbf{y}$$

Recap thus far!



$$\mathbf{A} = \mathbf{T}\hat{\mathbf{X}}^{-1}$$

A

$$\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$$

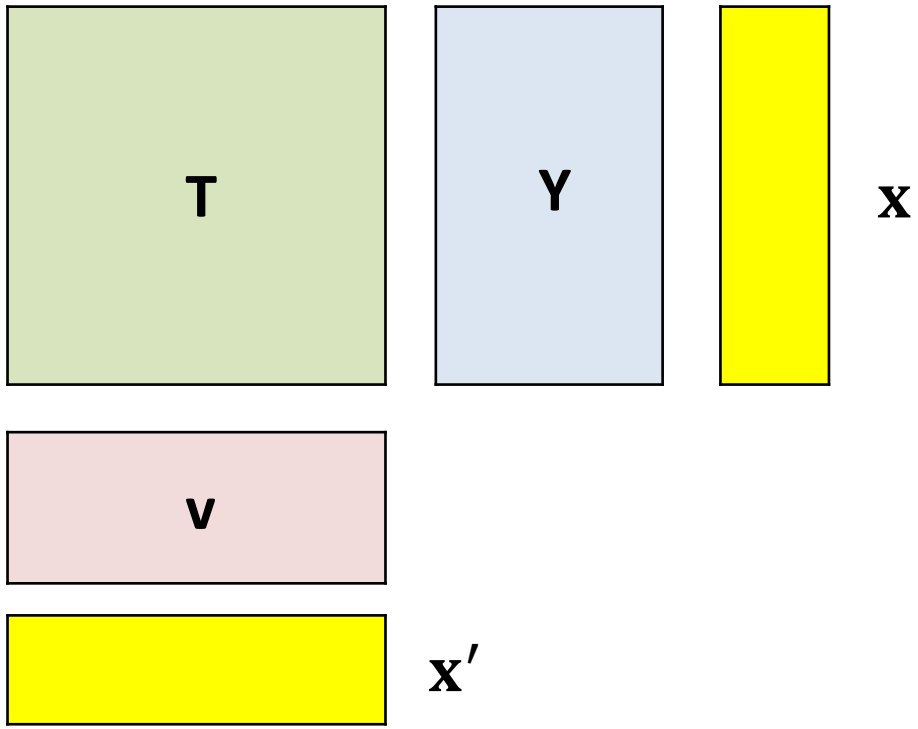
- Create an identity matrix (same size as the **A** matrix)
- Calculate **I – A**
- Calculate the inverse of **I – A** using MINVERSE

- **T** matrix; **A** matrix; **L** matrix – SAME SIZE!
- **A** matrix – Direct requirements matrix
- **L** matrix – Total requirements matrix

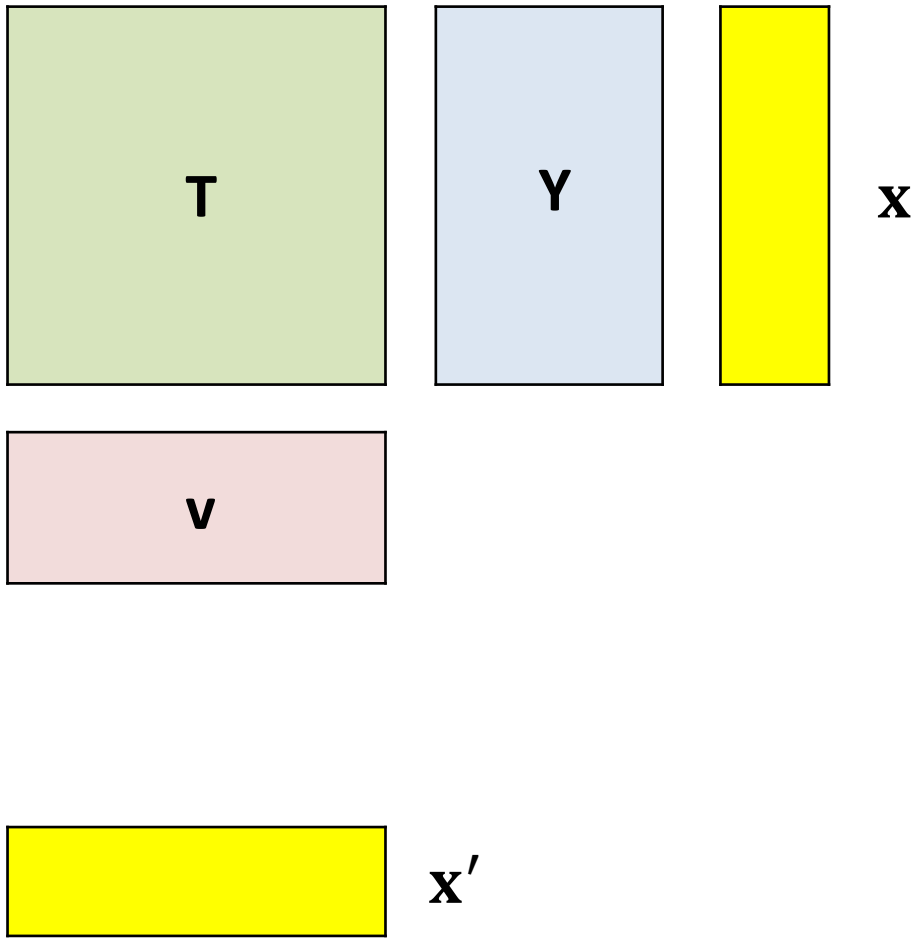
Let's say you want to carry out a carbon footprint assessment – you already have economic data (**T, Y, V** matrices).

How do we connect the data on emissions to the economic data in our IO table?

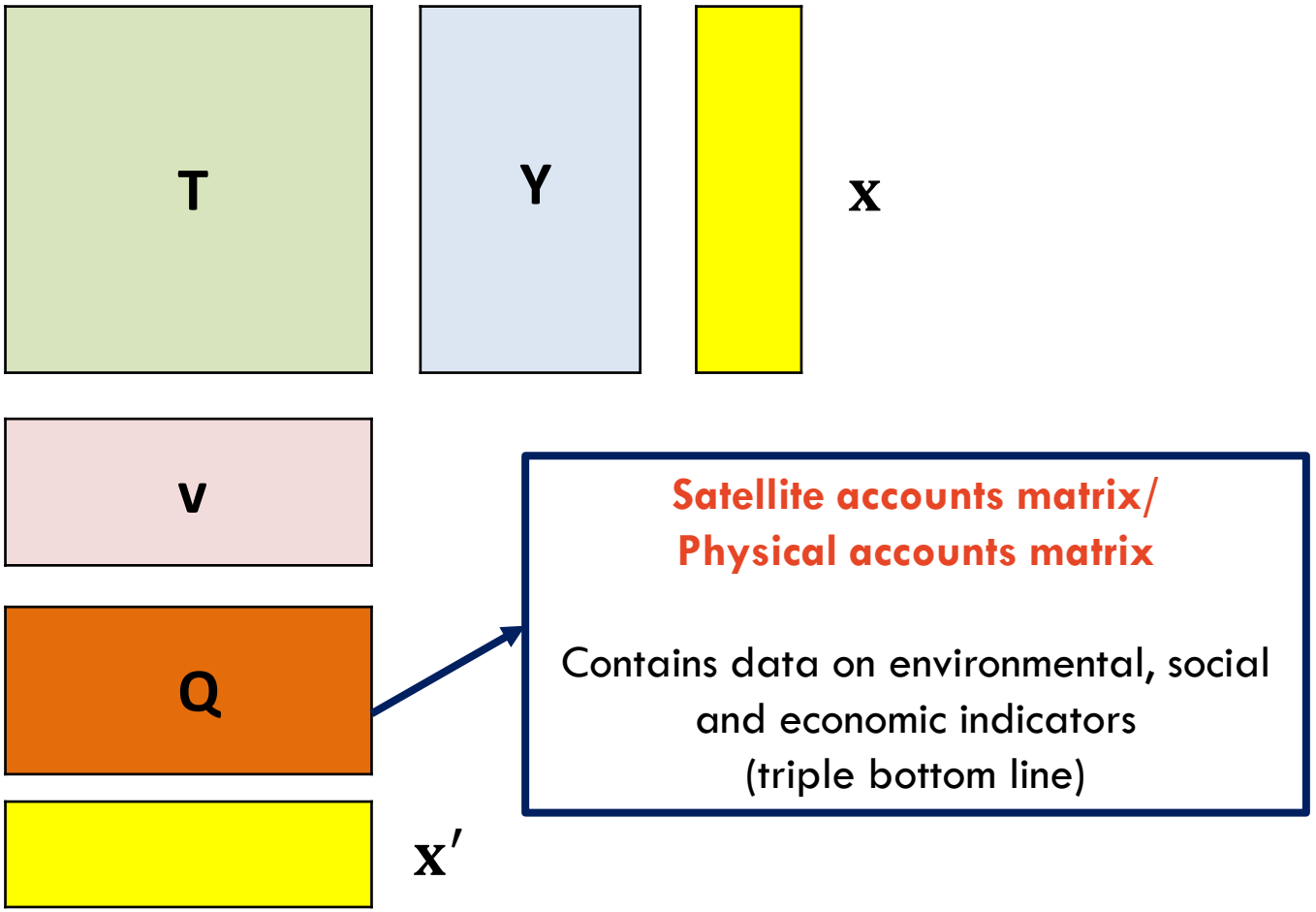
Economic input-output system



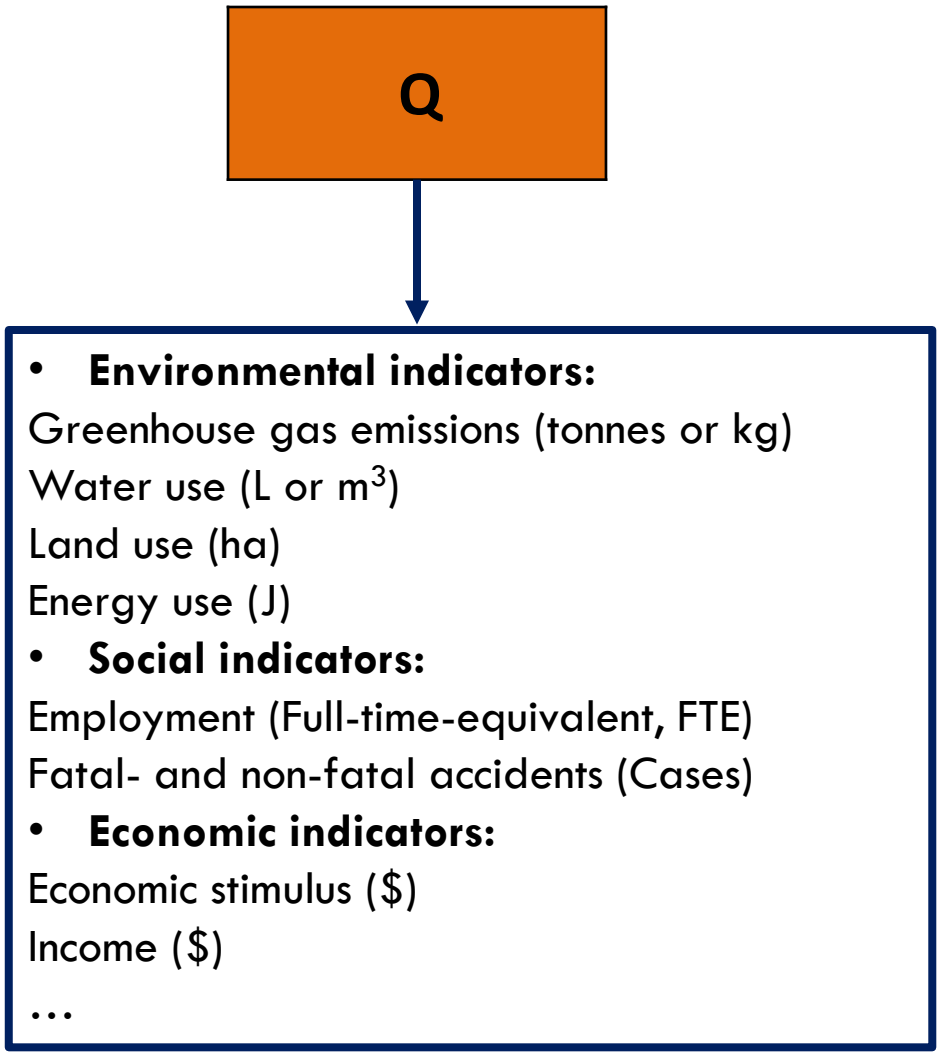
Economic input-output system



Satellite account matrix



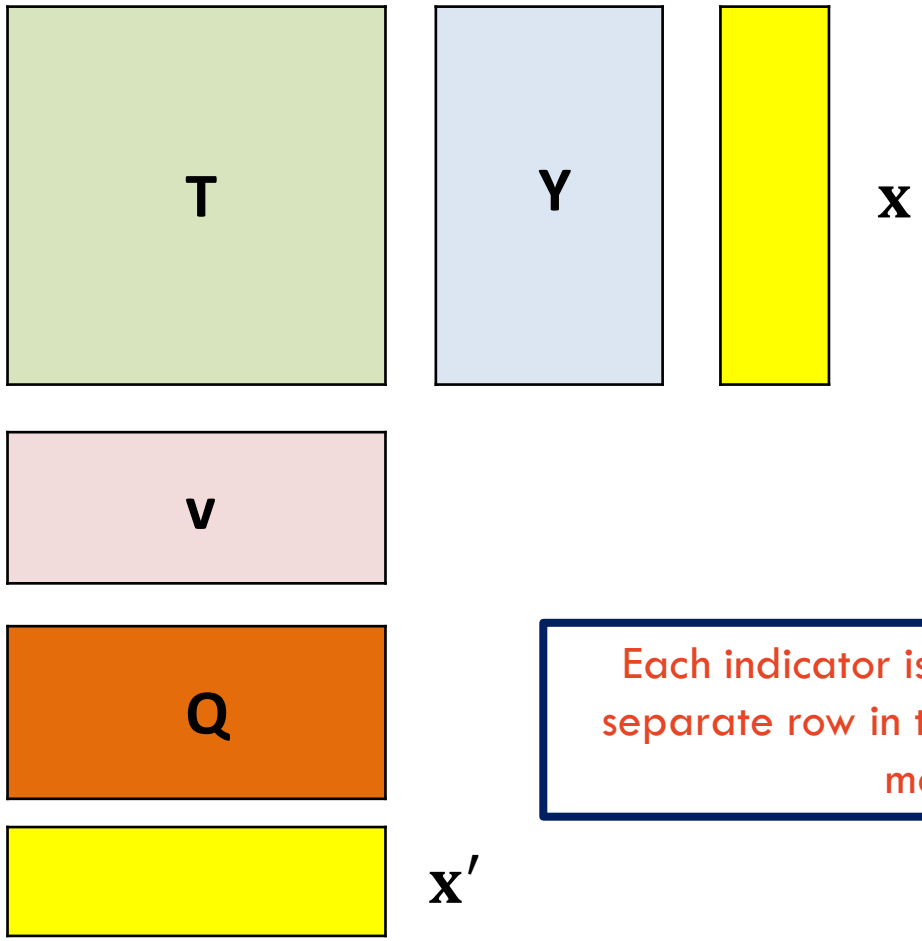
Satellite account matrix



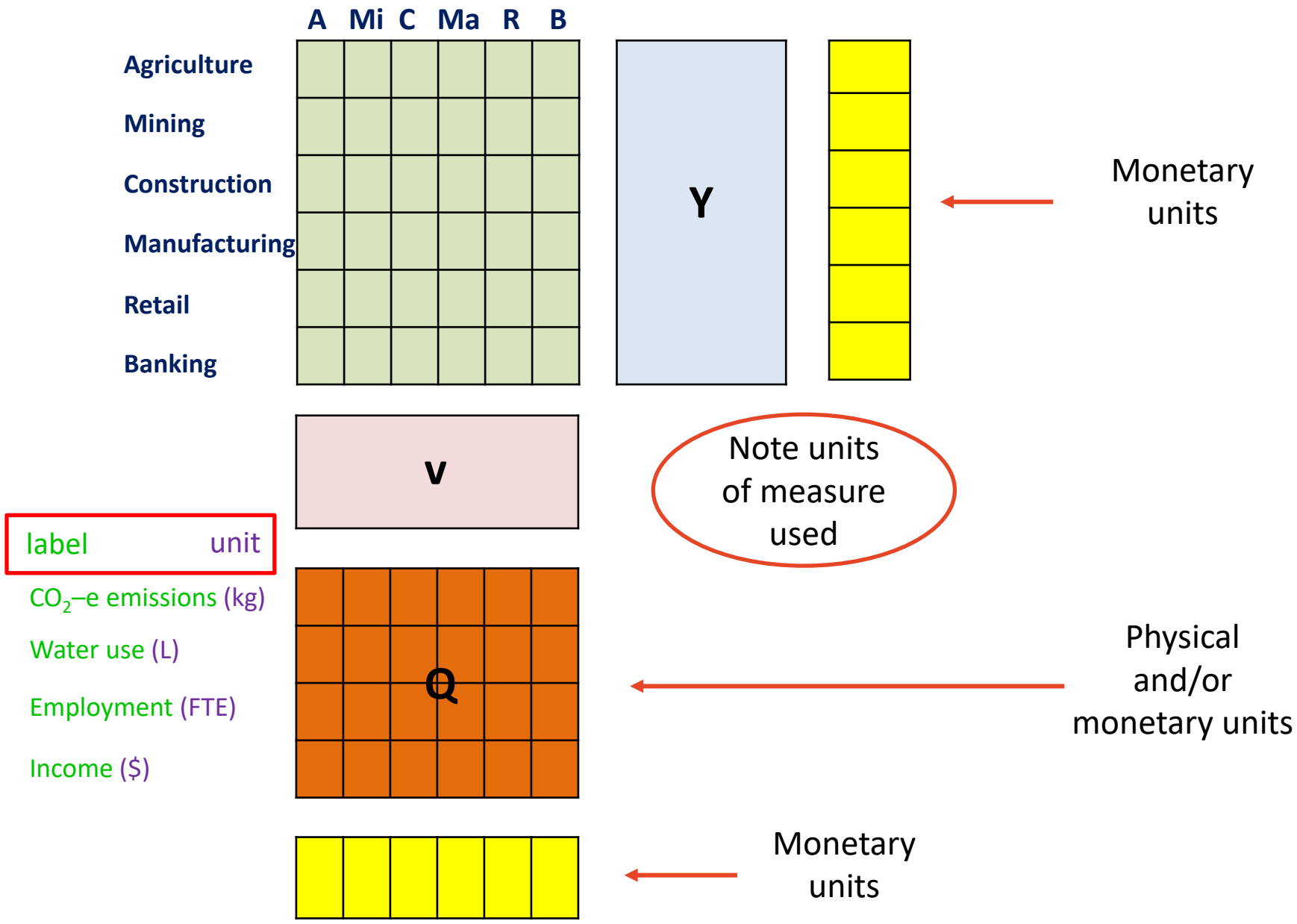
Satellite account matrix

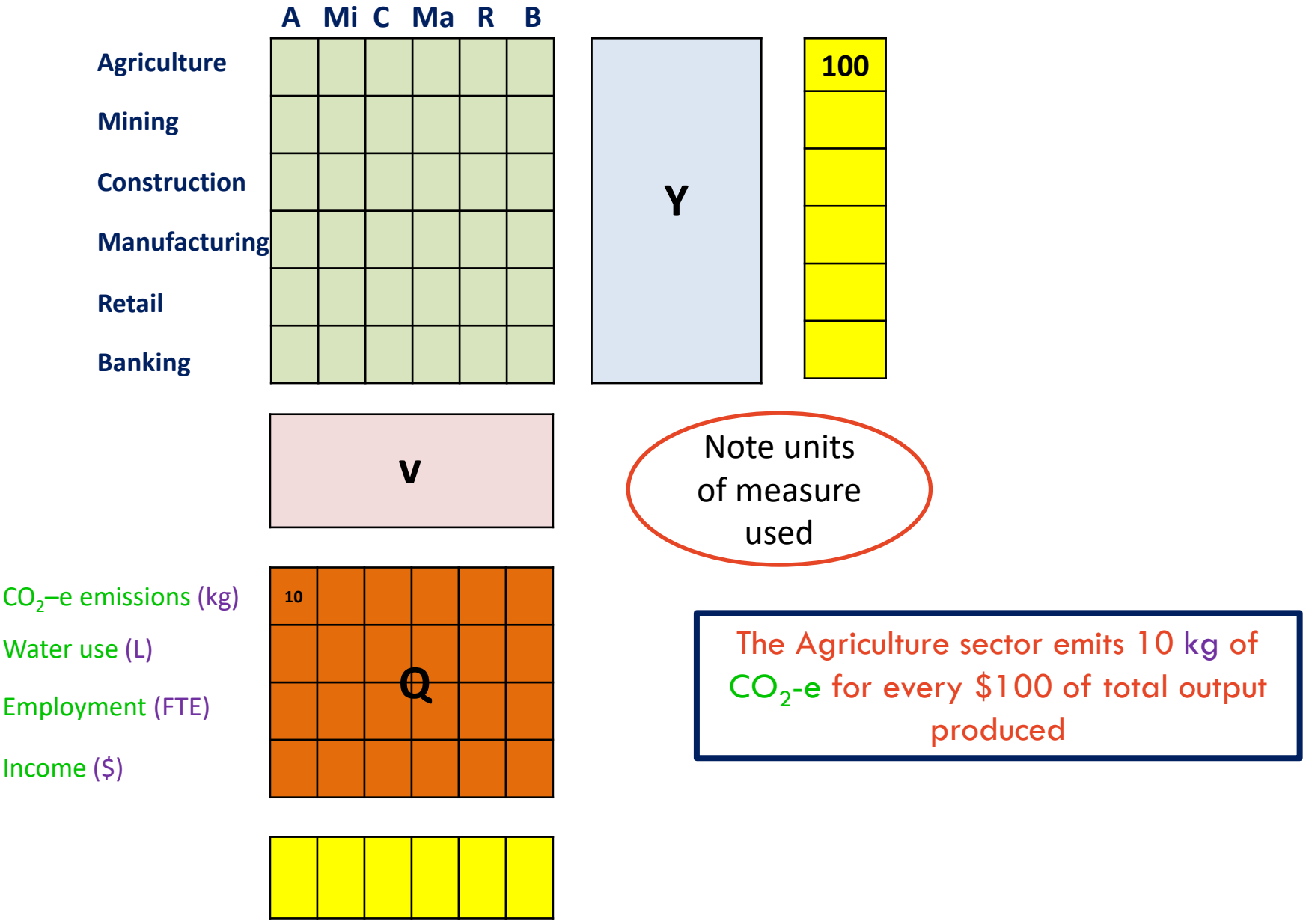


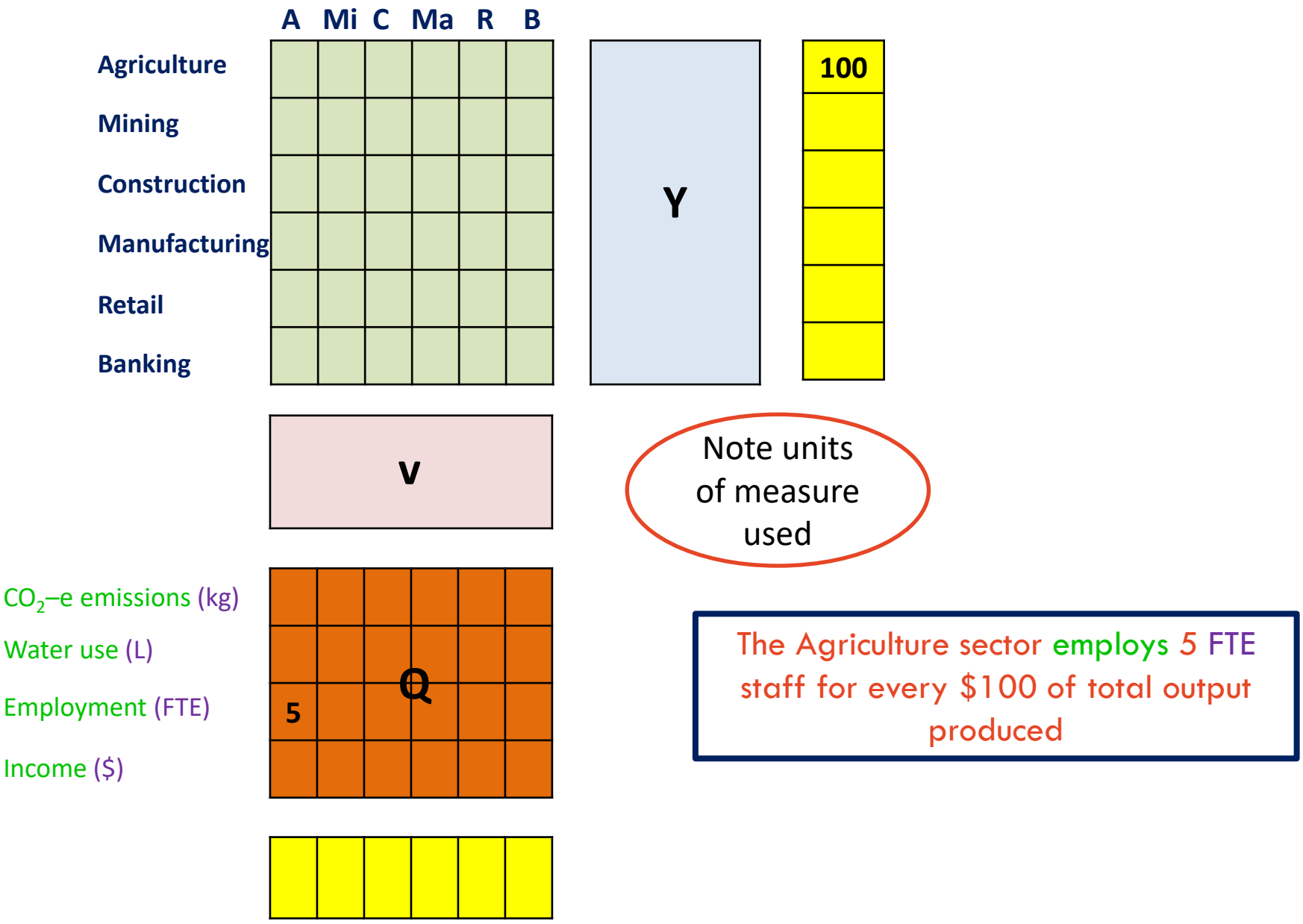
Satellite account matrix

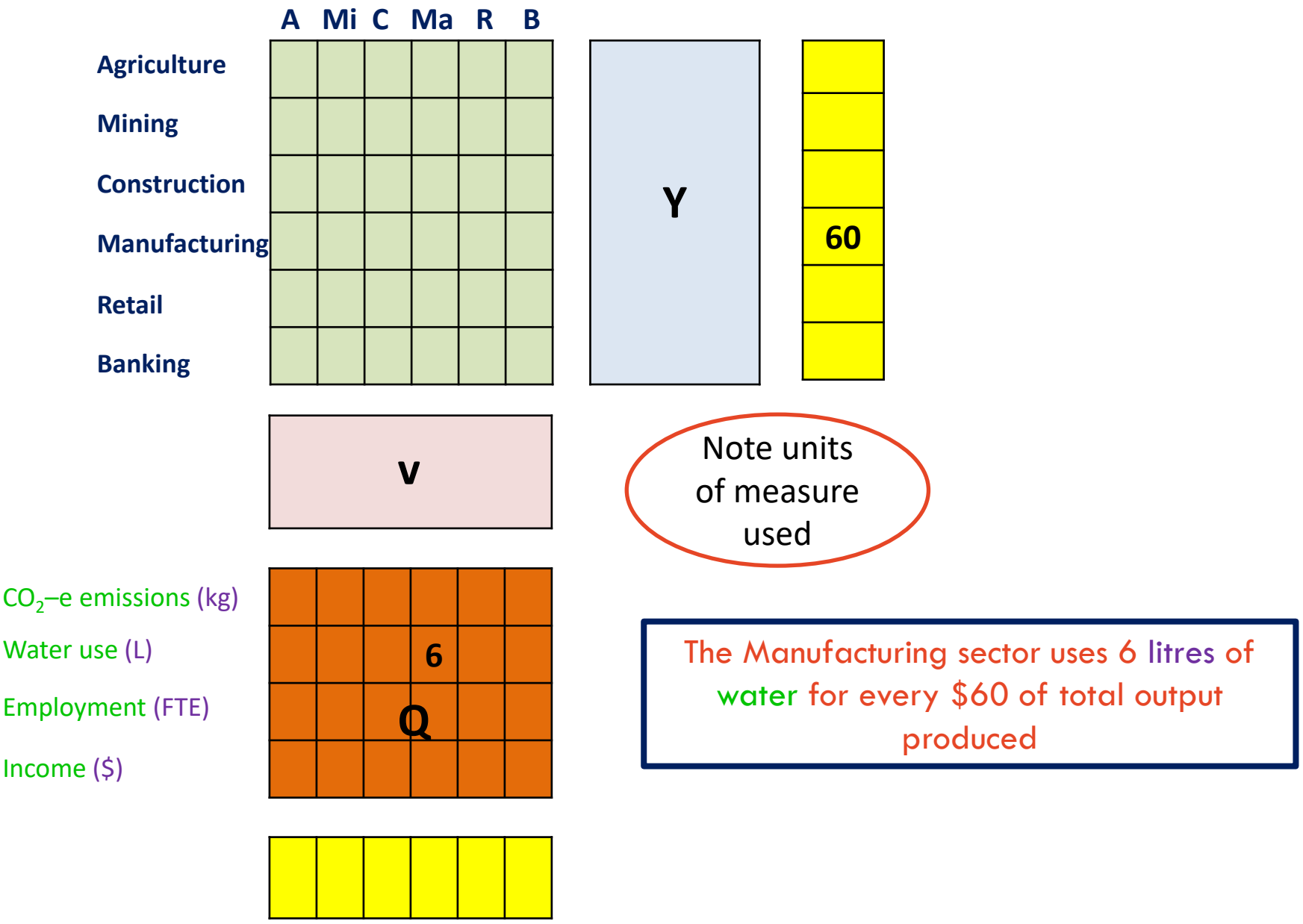


Each indicator is represented as a separate row in the satellite account matrix


A
Mi
C
Ma
R
B
Agriculture
Mining
Construction
Manufacturing
Retail
Banking
Y
Monetary units
v
Note units of measure used
label
unit
CO_2–e emissions (kg)
Water use (L)
Employment (FTE)
Income ($)
Q
Physical and/or monetary units
Monetary units


A
Mi
C
Ma
R
B
Agriculture
Mining
Construction
Manufacturing
Retail
Banking
Y
100
V
Note units of measure used
CO_2–e emissions (kg)
Water use (L)
Employment (FTE)
Income ($)
10
Q
The Agriculture sector emits 10 kg of CO_2-e for every $100 of total output produced


A
Mi
C
Ma
R
B
Agriculture
Mining
Construction
Manufacturing
Retail
Banking
Y
100
V
Note units of measure used
CO_2–e emissions (kg)
Water use (L)
Employment (FTE)
Income ($)
Q
5
The Agriculture sector employs 5 FTE staff for every $100 of total output produced


A
Mi
C
Ma
R
B
Agriculture
Mining
Construction
Manufacturing
Retail
Banking
Y
60
V
Note units of measure used
CO_2–e emissions (kg)
Water use (L)
Employment (FTE)
Income ($)
6
Q
The Manufacturing sector uses 6 litres of water for every $60 of total output produced

Value added

Satellite data: Australian Bureau of Statistics

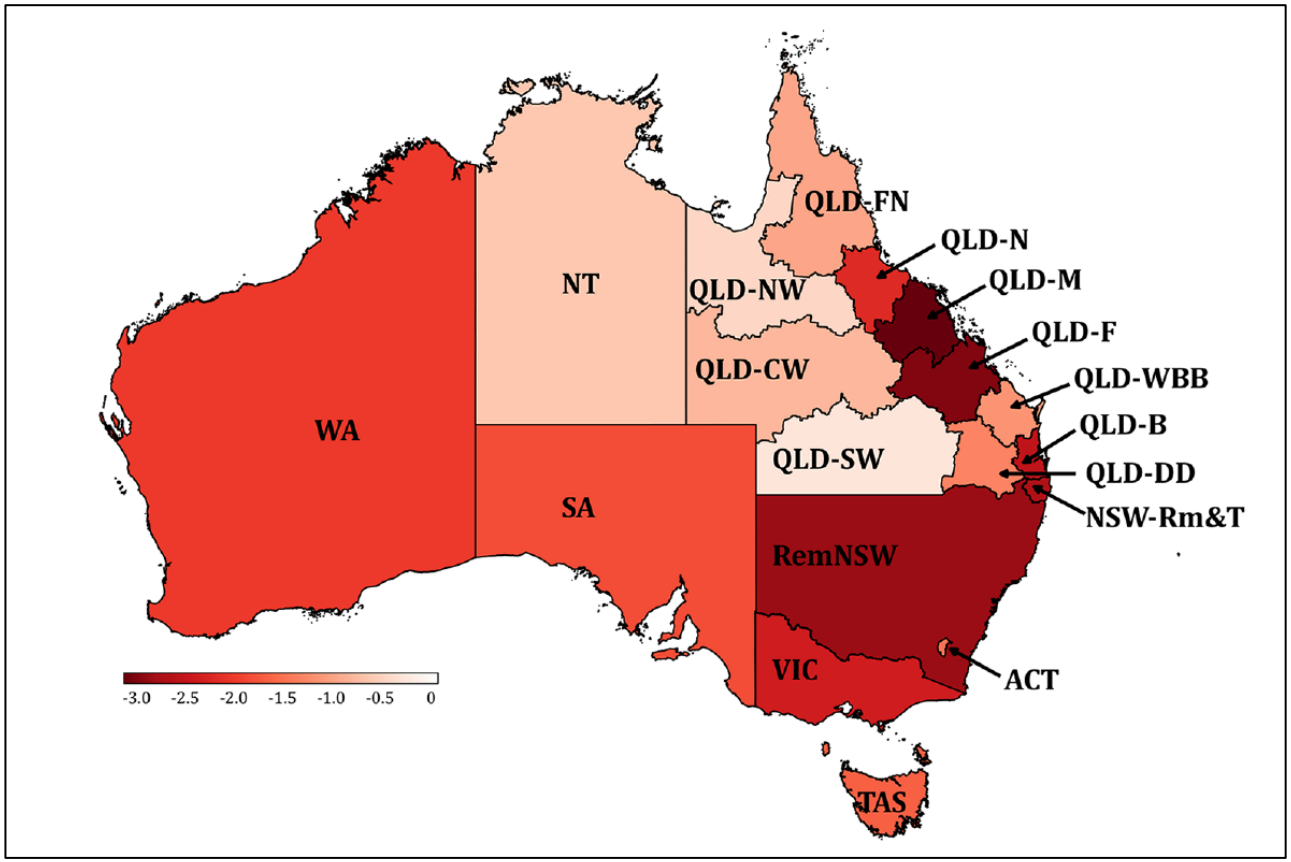


Figure 1. Geographical distribution of value-added loss caused by Tropical Cyclone Debbie.

Lenzen M, Malik A, Kenway S, Daniels P, Lam KL, Geschke A. Economic damage and spillovers from a tropical cyclone. Natural Hazards and Earth System Sciences. 2019; **19**(1): 137-51.

Emissions

Satellite data: National Greenhouse Gas Inventory

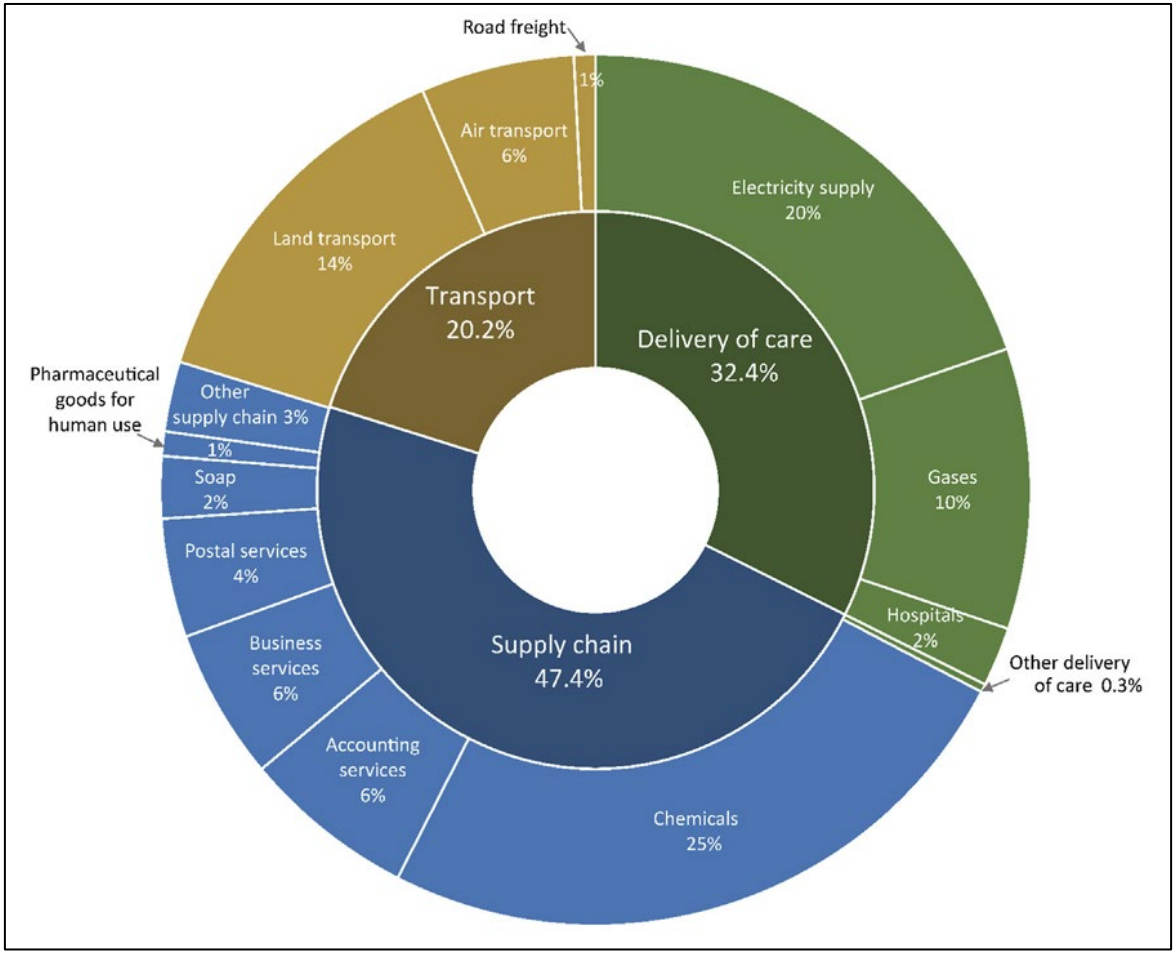


Figure 1. Breakdown of WA Health's carbon footprint by sector and by category. Sectors have been categorized into: delivery of care, which includes all emissions directly associated with the treatment of patients; supply chain, which includes all the upstream activity required to enable deliver of care; and transport.

Irwin, A, Malik, A, Vyas, A, Bateman, C & Joyce, S 2024, 'The carbon footprint of health care delivery in Western Australia's public health system', The Lancet regional health. Western Pacific, vol. 48.

Extinction risk

Satellite data: IUCN Red List of Threatened Species

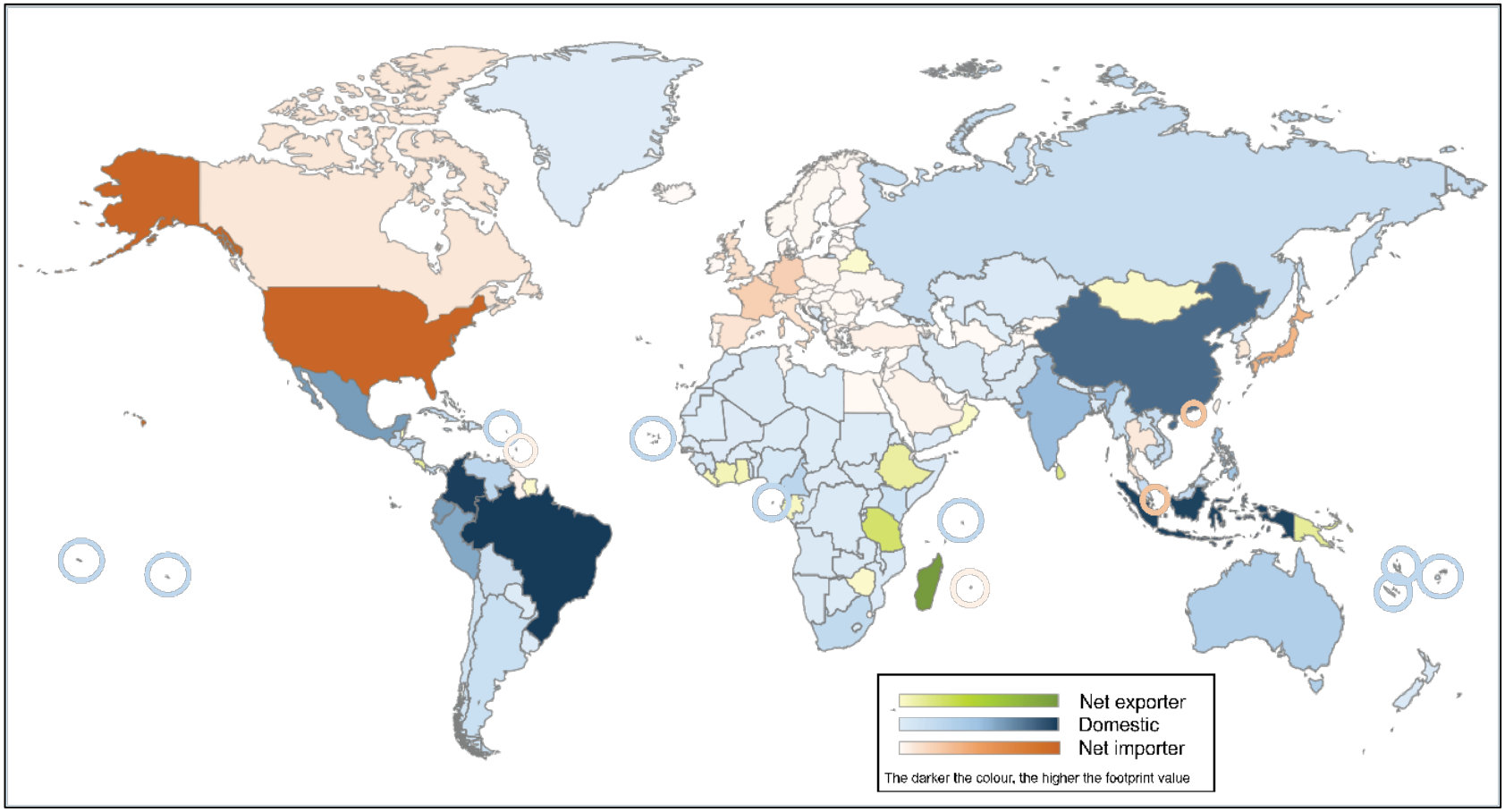


Figure 1. Category of extinction-risk footprint by country. Each of the 188 countries in scope is colour-coded according to which of the imported, exported, or domestic footprints has the highest value in that country. Net importers (orange) primarily drive extinction-risk footprint in other countries, the extinction-risk footprint for net exporters (green) is primarily driven by consumption in other countries, and consumption within the country primarily drives extinction-risk footprint for domestic countries (blue). The darker the colour, the higher the corresponding footprint value

Irwin, A, Geschke, A, Brooks, TM, Siikamäki, J, Mair, L & Strassburg, BBN 2022, 'Quantifying and categorising national extinction-risk footprints', *Scientific Reports,* vol. 12, no. 1, pp. 5861.

Nitrogen

Satellite data: United Nations Food and Agriculture Organisation and the International Fertilizer Industry Association

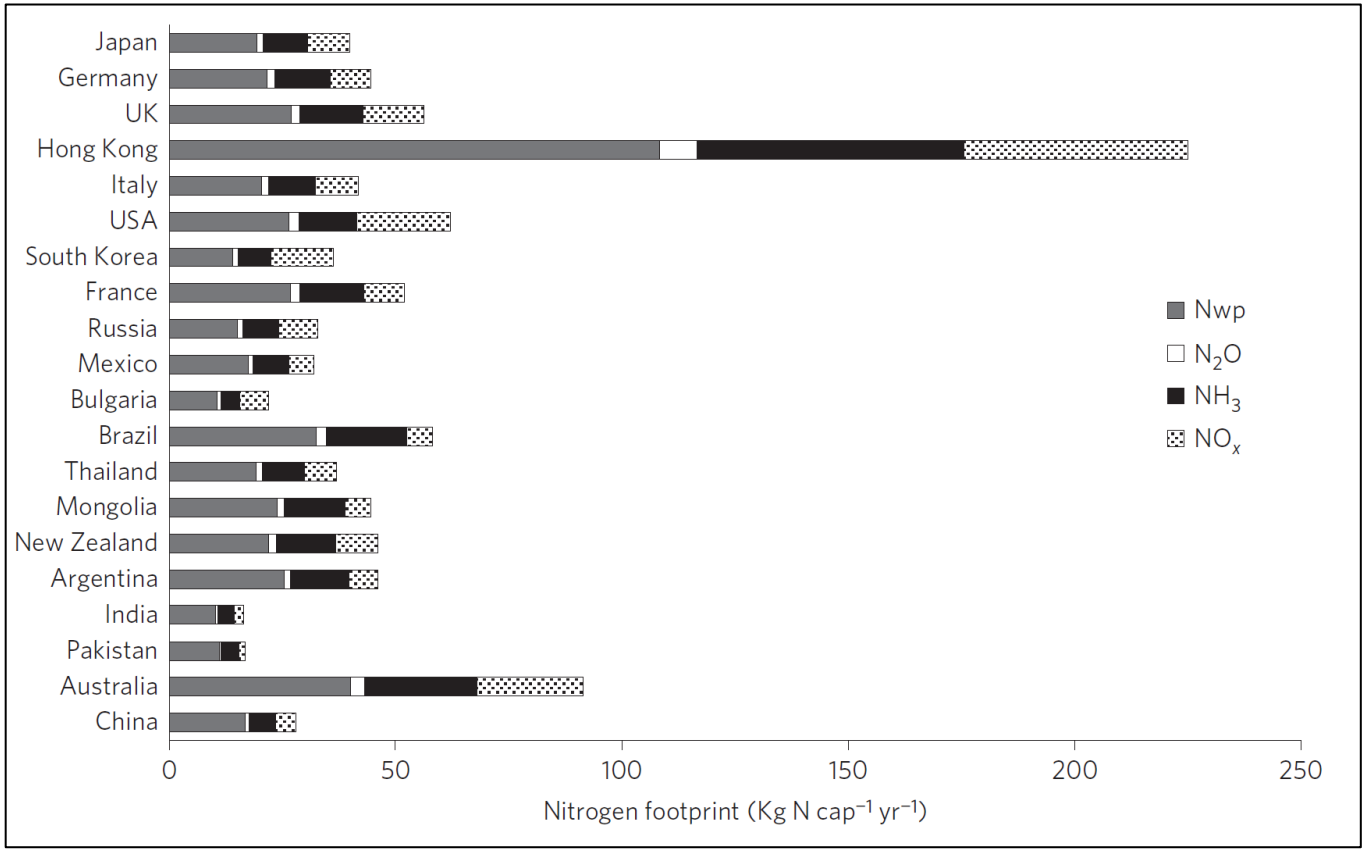


Figure 1b. Per-capita nitrogen footprints. Nwp: nitrogen potentially exportable to water bodies, mostly NO_3^-. Per-capita values include both emissions induced domestically and abroad from industries (including agriculture) and households for the nation's consumption

Oita, A., Malik, A., Kanemoto, K., Geschke, A., Nishijima, S. and Lenzen, M. (2016), "Substantial nitrogen pollution embedded in international trade ", *Nature Geoscience,*, Vol. 9 No. 2, pp. 111-115.

Global textiles industry employment

Satellite data: International Labour Organisation

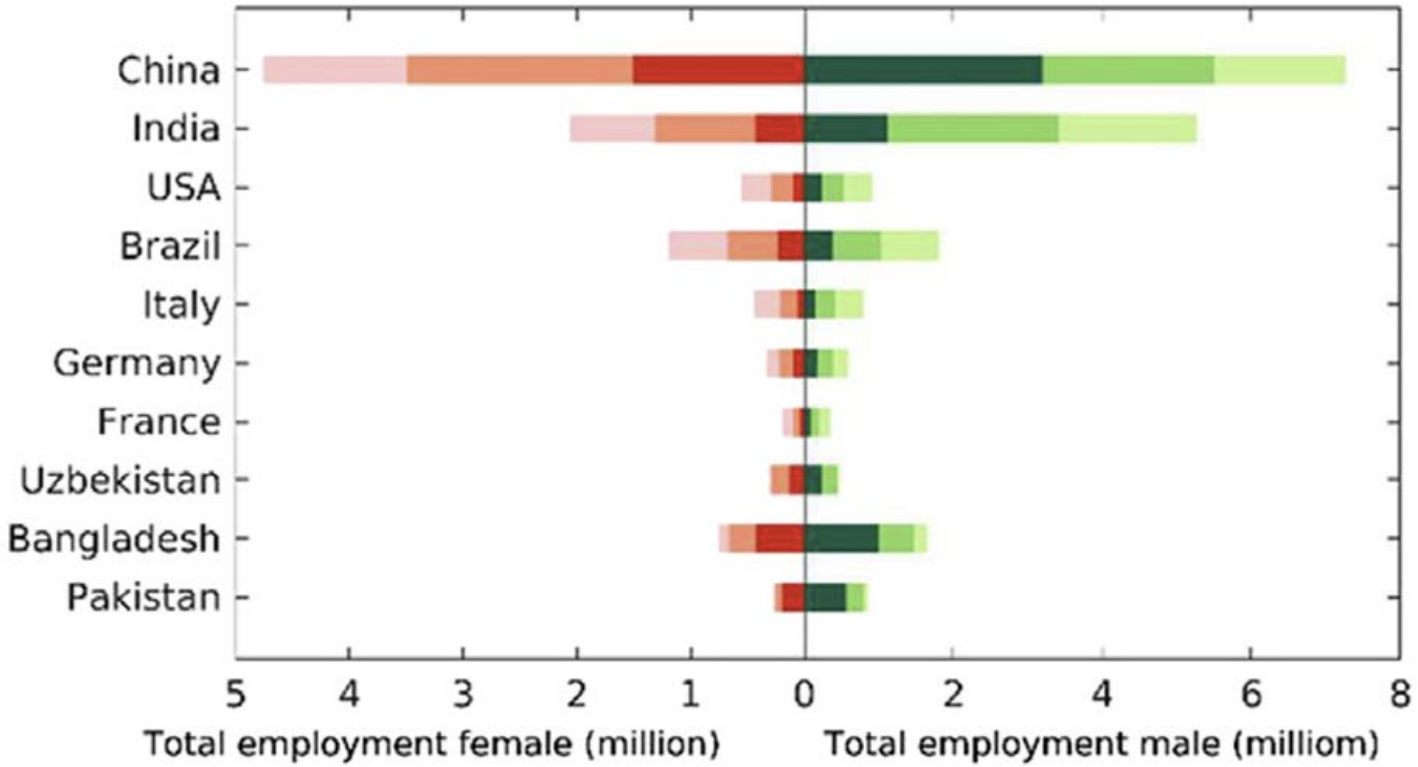


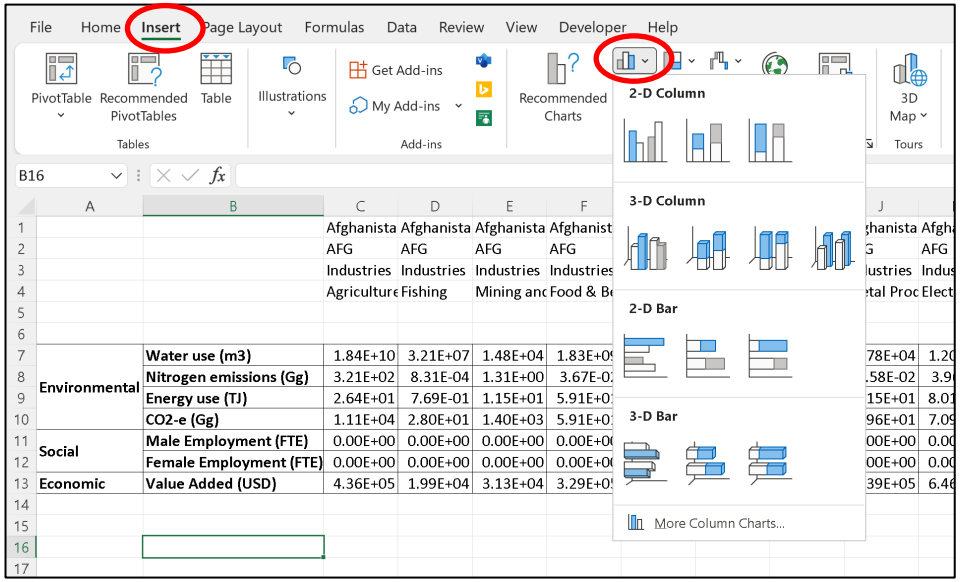
Fig. 3. Top 10 countries by textiles related total [...] female employment and male employment.

The darkest shade represents direct employment in the textiles industry, the medium shade represents employment generated in the direct supply chains of the textiles industry, and the lightest shade represents employment in all higher order supply chains.

Peters, G, Li, M & Lenzen, M 2021, 'The need to decelerate fast fashion in a hot climate - A global sustainability perspective on the garment industry', Journal of cleaner production, vol. 295.

Exploring global data - I

- Open Satellite Data file (see Canvas page for this week)
- Pick a country
- Plot one indicator, as a bar graph
- Make observations – which sectors are responsible for the key impacts?



How can we link raw data on indicators with an input-output table?

We need to construct the **Q** matrix

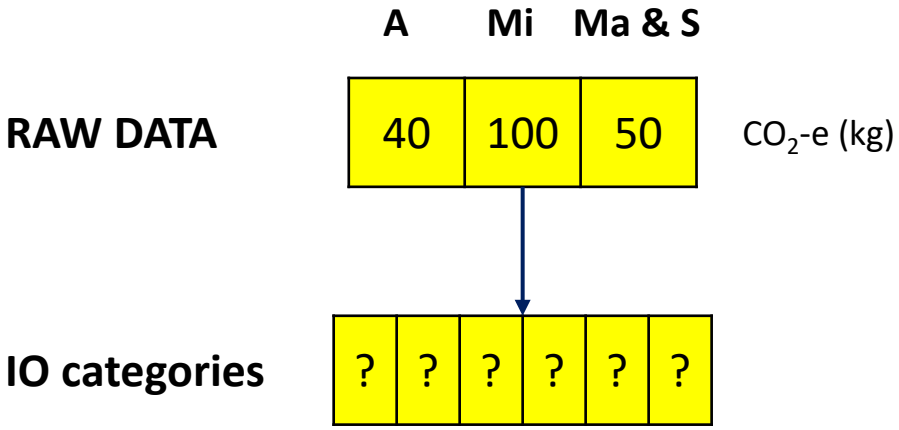
Steps for Q matrix construction

- Identify the sectoral structure of the 'raw' data for the indicator in question
- Create a concordance matrix to link the sectoral structure of the 'raw' data and the sectoral structure of the IO table
- Calculate a 'map' using this concordance matrix and a suitable proxy
- Multiply the 'raw' data by the calculated map to generate your Q matrix, which will have the same sectoral structure as your input-output table

Steps for Q matrix construction

PREPARATION

1) Get data on carbon dioxide emissions (e.g https://www.greenhouseaccounts.climatechange.gov.au/)
2) Examine the data – are the sectors used to present the raw data exactly the same as the input-output sectors? Often there is a mismatch!



A – Agriculture
Mi – Mining
Ma & S – Manufacturing and Services

PREPAR

1) Get
2) Exan
input

ange.gov.au/)

ame as the

Emissions for:

Location: AUSTRALIA

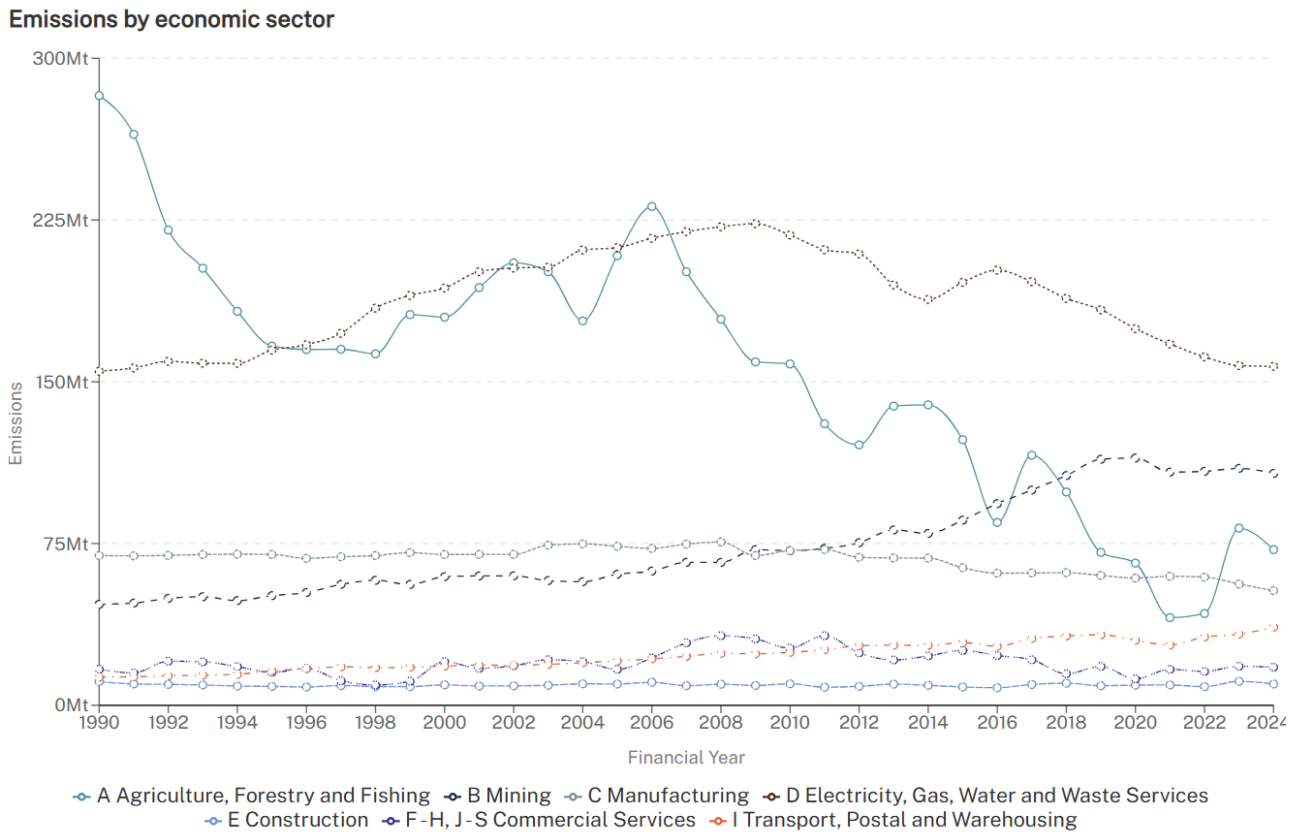
Sector(s): A Agriculture, Forestry and Fishing, B Mining, C Manufacturing, D Electricity, Gas, Water and Waste Services, E Construction, F-H, J-S Commercial Services, I Transport, Postal and Warehousing

Gas: Carbon Dioxide Equivalent - AR5

Graph | Table

Download data

View as: Bar Line



and Services

Steps for Q matrix construction

3) Construct a concordance matrix **C** – a binary matrix containing 0s and 1s. This documents where there is a hierarchical relationship between the sectors in the rows (the source data structure) and the sectors in the columns (the IO table structure)

Source data sectors →	IO table sectors → Agriculture	Mining	Construction	Manufacturing	Retail	Banking
A	**1**	**0**	**0**	**0**	**0**	**0**
Mi	**0**	**1**	**0**	**0**	**0**	**0**
Ma & S	**0**	**0**	**1**	**1**	**1**	**1**

Construction is a sub-sector of the Manufacturing & Services sector

Manufacturing is a sub-sector of the Manufacturing & Services sector

Retail is a sub-sector of the Manufacturing & Services sector

Banking is a sub-sector of the Manufacturing & Services sector

A – Agriculture
Mi – Mining
Ma & S – Manufacturing and Services

Steps for Q matrix construction

3) Construct a concordance matrix **C** – binary matrix containing 0s and 1s.

	Agriculture	Mining	Construction	Manufacturing	Retail	Banking
A	**1**	**0**	**0**	**0**	**0**	**0**
Mi	**0**	**1**	**0**	**0**	**0**	**0**
Ma & S	**0**	**0**	**1**	**1**	**1**	**1**

CO_2-e (kg)

RAW DATA

40
100
50

Question: Can you multiply the raw data vector with the concordance matrix to get satellite data in the IO-classification?

RAW DATA

40
100
50

Total = 190

CO_2-e (kg)

IO categories

?	?	?	?	?	?

40	100	50	50	50	50

Total = 340

CO_2-e (kg)

Steps for Q matrix construction

3) Construct a concordance matrix **C** – binary matrix containing 0s and 1s.

	Agriculture	Mining	Construction	Manufacturing	Retail	Banking
A	**1**	**0**	**0**	**0**	**0**	**0**
Mi	**0**	**1**	**0**	**0**	**0**	**0**
Ma & S	**0**	**0**	**1**	**1**	**1**	**1**

- Need to find ratios so that when you add all values in any row of **C**, you get 1
- Ratios can be calculated by using a proxy – typically total output is used

4) Calculate the total output vector (**x**) of the IO table

Total Output AUD$
20538
15152
24384
27614
10162
13599

Steps for Q matrix construction

5) Calculate $\mathbf{Cx}$ (using matrix multiplication)

For the worked example, this step will yield a 3 x 1 vector

20538
15152
75759

6) Calculate $\widehat{\mathbf{Cx}}$

Turn $\mathbf{Cx}$ into a diagonal matrix

For the worked example, this step will yield a 3 x 3 matrix

20538	0	0
0	15152	0
0	0	75759

7) Calculate $\widehat{\mathbf{Cx}}^{-1}$

Calculate the inverse of $\widehat{\mathbf{Cx}}$

For the worked example, this step will yield a 3 x 3 matrix

0.00004869	0	0
0	0.00006600	0
0	0	0.00001320

8) Calculate $\widehat{\mathbf{X}}$

Turn $\mathbf{x}$ into a diagonal matrix

For the worked example, this step will yield a 6 x 6 matrix

20538	0	0	0	0	0
0	15152	0	0	0	0
0	0	24384	0	0	0
0	0	0	27614	0	0
0	0	0	0	10162	0
0	0	0	0	0	13599

9) Calculate $\mathbf{C}\widehat{\mathbf{X}}$ (using matrix multiplication)

Multiply the concordance matrix with the diagonal $\mathbf{x}$ matrix

For the worked example, this step will yield a 3 x 6 matrix

20538	0	0	0	0	0
0	15152	0	0	0	0
0	0	24384	27614	10162	13599

10) Calculate the map $\widehat{\mathbf{Cx}}^{-1} * \mathbf{C}\widehat{\mathbf{X}}$ (using matrix multiplication)

Multiply the answer from Step 7 with the answer from Step 9 to get the Map. For the worked example, this step will yield a 3 x 6 matrix.

1.00	0	0	0	0	0
0	1.00	0	0	0	0
0	0	0.32	0.36	0.13	0.18

Steps for Q matrix construction

[Compare the concordance matrix with its associated map]

Concordance

	Agriculture	Mining	Construction	Manufacturing	Retail	Banking
A	1	0	0	0	0	0
Mi	0	1	0	0	0	0
Ma & S	0	0	1	1	1	1

Map

	Agriculture	Mining	Construction	Manufacturing	Retail	Banking
A	1	0	0	0	0	0
Mi	0	1	0	0	0	0
Ma & S	0	0	0.32	0.36	0.13	0.18

Steps for Q matrix construction

11) Multiply the raw data vector with the map (using matrix multiplication)

Raw data

CO_2-e (kg)

A	Mi	Ma & S
40	100	50

Total = 190

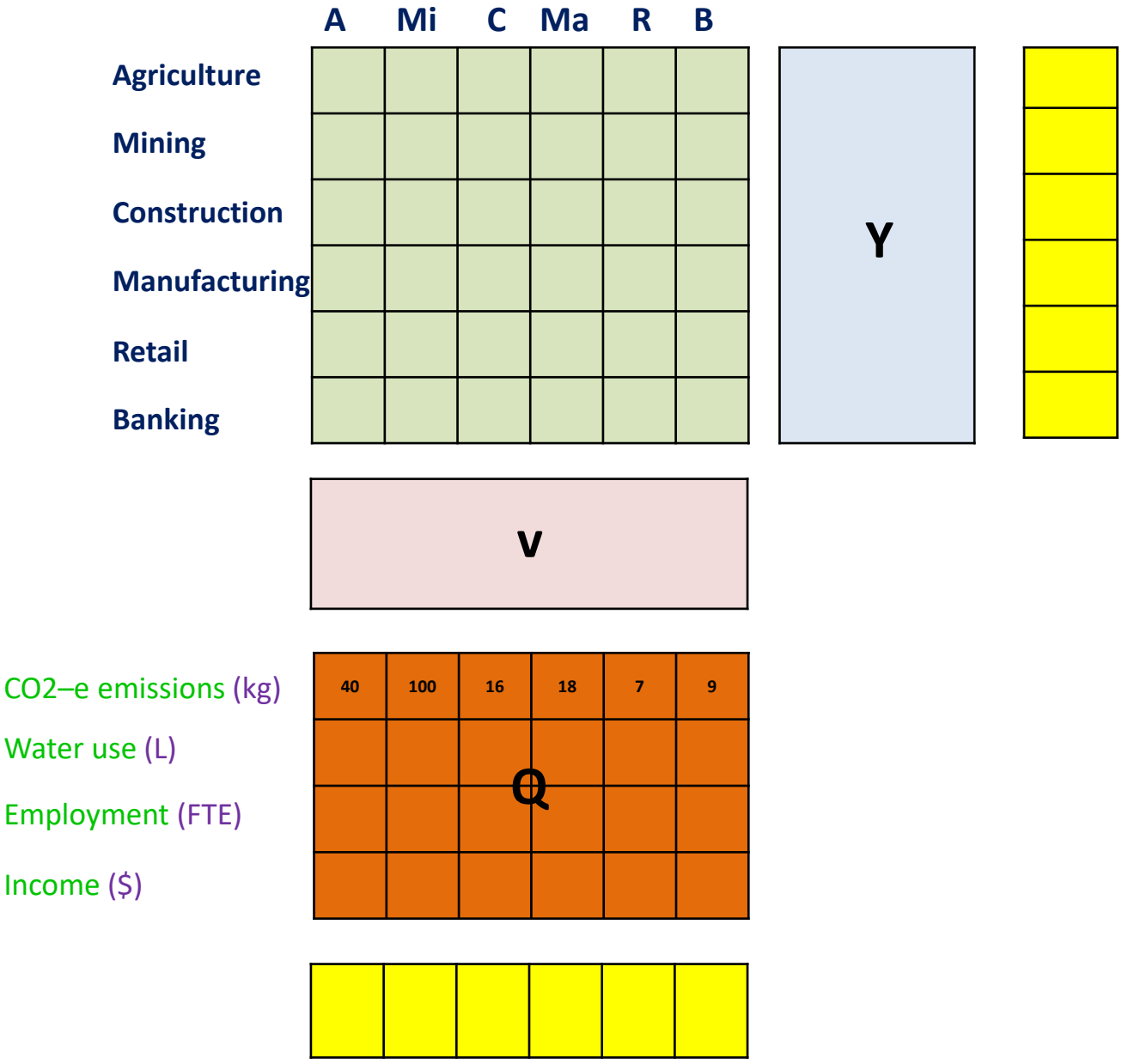
	Agriculture	Mining	Construction	Manufacturing	Retail	Banking
A	1	0	0	0	0	0
Mi	0	1	0	0	0	0
Ma & S	0	0	0.32	0.36	0.13	0.18

Size check (1x3) x (3x6) = (1x6)

Final Q matrix

CO_2-e (kg)	40	100	16	18	7	9

Total = 190


A
Mi
C
Ma
R
B
Agriculture
Mining
Construction
Manufacturing
Retail
Banking
Y
V
CO2–e emissions (kg)
Water use (L)
Employment (FTE)
Income ($)
40
100
16
18
7
9
Q

Tips for concordance construction #1

	Agriculture	Mining	Construction	Manufacturing	Retail	Banking	
A	1	0	0	0	0	0	=1
Mi	0	1	0	0	0	0	=1
Ma & S	0	0	1	1	1	1	=4
	=1	=1	=1	=1	=1	=1	

Concordance

sum down each column = 1

Map

sum across each row = 1

	Agriculture	Mining	Construction	Manufacturing	Retail	Banking	
A	1	0	0	0	0	0	=1
Mi	0	1	0	0	0	0	=1
Ma & S	0	0	0.32	0.36	0.13	0.18	=1
	=1	=1	=0.32	=0.36	=0.13	=0.18	

Tips for concordance construction #2

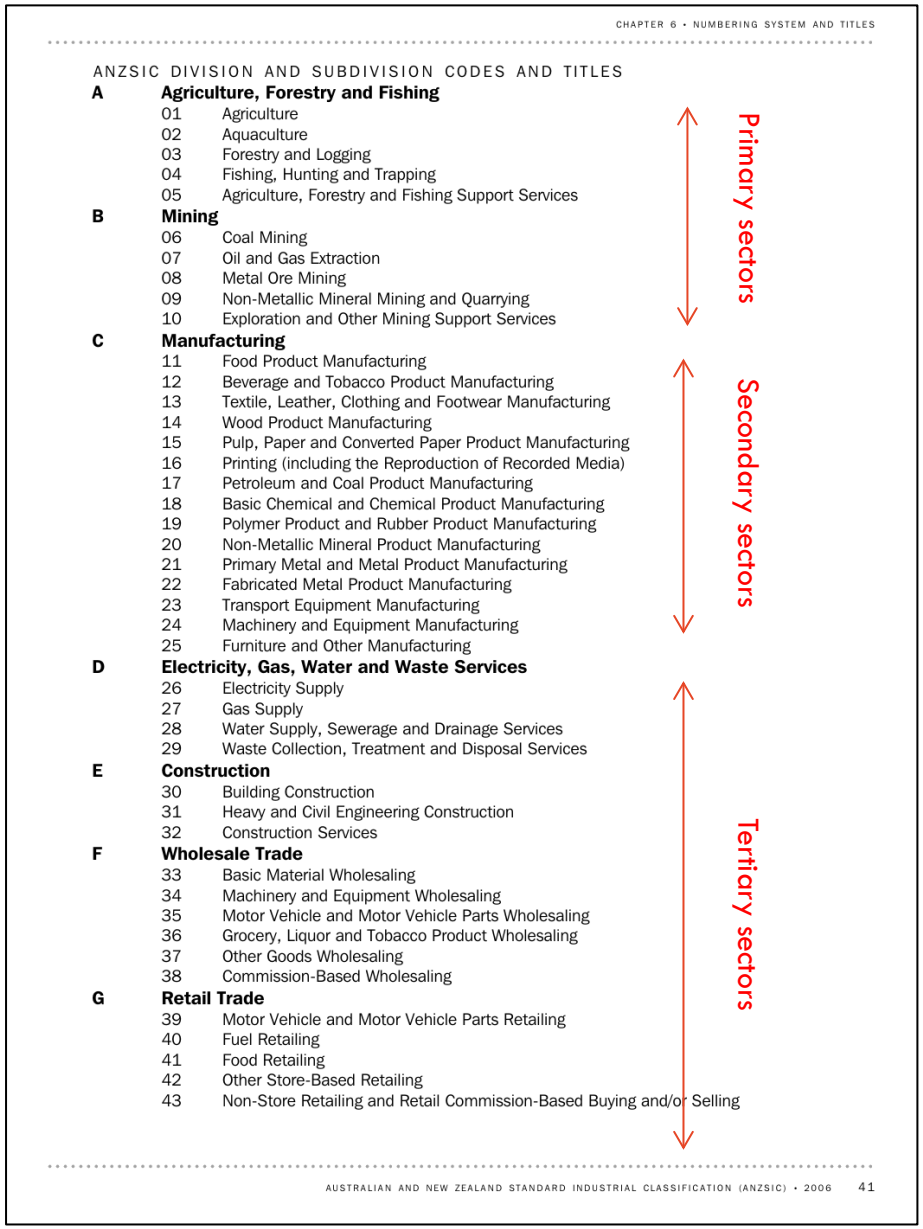
	Agriculture	Mining	Construction	Manufacturing	Retial	Banking
A	0	0	1	0	0	0
M	1	1	0	0	0	0
Ma & S	0	0	0	1	1	1

Check that the connections you have documented make sense in relation to the hierarchy of sectors

- Is Construction a sub-sector of the Agriculture sector?
- Is Agriculture a sub-sector of the Mining sector?

	Agriculture	Mining	Construction	Manufacturing	Retial	Banking
A	1	0	0	0	0	0
M	0	1	0	0	0	0
Ma & S	0	0	1	1	1	1

Tips for concordance construction #3



CHAPTER 6 • NUMBERING SYSTEM AND TITLES

ANZSIC DIVISION AND SUBDIVISION CODES AND TITLES

A	**Agriculture, Forestry and Fishing**	
	01	Agriculture
	02	Aquaculture
	03	Forestry and Logging
	04	Fishing, Hunting and Trapping
	05	Agriculture, Forestry and Fishing Support Services
B	**Mining**	
	06	Coal Mining
	07	Oil and Gas Extraction
	08	Metal Ore Mining
	09	Non-Metallic Mineral Mining and Quarrying
	10	Exploration and Other Mining Support Services
C	**Manufacturing**	
	11	Food Product Manufacturing
	12	Beverage and Tobacco Product Manufacturing
	13	Textile, Leather, Clothing and Footwear Manufacturing
	14	Wood Product Manufacturing
	15	Pulp, Paper and Converted Paper Product Manufacturing
	16	Printing (including the Reproduction of Recorded Media)
	17	Petroleum and Coal Product Manufacturing
	18	Basic Chemical and Chemical Product Manufacturing
	19	Polymer Product and Rubber Product Manufacturing
	20	Non-Metallic Mineral Product Manufacturing
	21	Primary Metal and Metal Product Manufacturing
	22	Fabricated Metal Product Manufacturing
	23	Transport Equipment Manufacturing
	24	Machinery and Equipment Manufacturing
	25	Furniture and Other Manufacturing
D	**Electricity, Gas, Water and Waste Services**	
	26	Electricity Supply
	27	Gas Supply
	28	Water Supply, Sewerage and Drainage Services
	29	Waste Collection, Treatment and Disposal Services
E	**Construction**	
	30	Building Construction
	31	Heavy and Civil Engineering Construction
	32	Construction Services
F	**Wholesale Trade**	
	33	Basic Material Wholesaling
	34	Machinery and Equipment Wholesaling
	35	Motor Vehicle and Motor Vehicle Parts Wholesaling
	36	Grocery, Liquor and Tobacco Product Wholesaling
	37	Other Goods Wholesaling
	38	Commission-Based Wholesaling
G	**Retail Trade**	
	39	Motor Vehicle and Motor Vehicle Parts Retailing
	40	Fuel Retailing
	41	Food Retailing
	42	Other Store-Based Retailing
	43	Non-Store Retailing and Retail Commission-Based Buying and/or Selling

Primary sectors

Secondary sectors

Tertiary sectors

AUSTRALIAN AND NEW ZEALAND STANDARD INDUSTRIAL CLASSIFICATION (ANZSIC) • 2006 41

Sector classification hierarchy
≠
economic interactions between sectors

Use sector classification scheme documentation to help understand the hierarchy of sectors in the economy,

e.g. Australia & New Zealand: ANZSIC 2006

Global: UN Central Product Classification Scheme

Tutorial



THE UNIVERSITY OF
SYDNEY

Tutorial exercise

Week 5 Activities

Consider the following input-output matrix.

($100million)	Sheep, Grain, Beef and Dairy Cattle	Poultry and Other Livestock	Coal Mining	Beer Manufacturing	Textile Manufacturing	Chemical Manufacturing	Retail Trade	Technical Services	Banking	Defence		Households	Government	Exports
Sheep, Grain, Beef and Dairy Cattle	40	30	1	40	2	4	1	1	0	0		3	10	72
Poultry and Other Livestock	35	25	0	2	10	10	5	1	0	0		20	5	48
Coal Mining	2	1	25	0	0	10	0	15	0	0		0	12	125
Beer Manufacturing	1	2	5	38	0	2	10	0	0	0		14	10	59
Textile Manufacturing	10	5	0	5	25	12	3	1	0	5		10	6	28
Chemical Manufacturing	22	25	42	8	9	24	1	2	0	32		5	15	15
Retail Trade	12	8	4	3	12	4	24	28	21	8		32	8	12
Technical Services	24	25	19	12	20	30	8	20	48	50		15	20	7
Banking	8	7	25	6	4	12	15	25	30	50		28	40	44
Defence	0	0	0	0	0	0	0	0	0	12		0	146	100
Households	12	3	15	5	6	10	55	85	63	10				
Government	10	9	13	4	5	10	15	12	20	52				
Imports	28	21	41	18	17	72	39	108	112	39				

a) Suppose you have data on **CO_2 emissions** for 4 'raw data' sectors:

- Agriculture, Forestry and Fishing (1,000 tonnes)
- Mining (5,000,000 tonnes)
- Manufacturing (32,000 tonnes)
- Services (950 tonnes)

Construct a CO_2 satellite account for the input-output table.

Tutorial exercise

b) Suppose you have data on a range of environmental indicators, each with a different number of 'raw data' sectors.

Energy data:

- Agriculture, Forestry and Fishing (250 MJ)
- Mining (1,000 MJ)
- Manufacturing (750 MJ)
- Computer Services (320 MJ)
- Other Services (190 MJ)

Water data:

- Agriculture (95,632 GL)
- Mining (25,360 GL)
- Manufacturing and services (50,000 GL)

Nitrogen emissions data:

- Agriculture (950 tonnes)
- Mining (590 tonnes)
- Beer Manufacturing (296 tonnes)
- Textile Manufacturing (548 tonnes)
- Chemical Manufacturing (657 tonnes)
- Services (125 tonnes)

Week 5 Activities

Consider the following input-output matrix.

($100million)	Sheep, Grain, Beef and Dairy Cattle	Poultry and Other Livestock	Coal Mining	Beer Manufacturing	Textile Manufacturing	Chemical Manufacturing	Retail Trade	Technical Services	Banking	Defence		Households	Government	Exports
Sheep, Grain, Beef and Dairy Cattle	40	30	1	40	2	4	1	1	0	0		3	10	72
Poultry and Other Livestock	35	25	0	2	10	10	5	1	0	0		20	5	48
Coal Mining	2	1	25	0	0	10	0	15	0	0		0	12	125
Beer Manufacturing	1	2	5	38	0	2	10	0	0	0		14	10	59
Textile Manufacturing	10	5	0	5	25	12	3	1	0	5		10	6	28
Chemical Manufacturing	22	25	42	8	9	24	1	2	0	32		5	15	15
Retail Trade	12	8	4	3	12	4	24	28	21	8		32	8	12
Technical Services	24	25	19	12	20	30	8	20	48	50		15	20	7
Banking	8	7	25	6	4	12	15	25	30	50		28	40	44
Defence	0	0	0	0	0	0	0	0	0	12		0	146	100
Households	12	3	15	5	6	10	55	85	63	10				
Government	10	9	13	4	5	10	15	12	20	52				
Imports	28	21	41	18	17	72	39	108	112	39				

Construct a satellite account containing four indicators – CO_2, water, energy and nitrogen. Make sure you save these results – we'll use these satellites next week.

Homework for Week 5

Homework week 5 (Due: September 9, 2026)

For your 14-sector input-output table, complete the following:

Use the Satellite Construction process outlined in the lecture to construct a Satellite matrix with three indicators (one environmental, one social and one economic) for your Input-Output table.

Ensure that the initial ('raw') sectors for your indicators **are different** from the sectors in your IO table. Build an appropriate concordance matrix for each indicator. Using the steps and techniques outlined in the lecture, use the total output of your IO table as the proxy to obtain a map and then construct a satellite matrix for your IO table.

For the economic indicator, you can simply extract the first row of the value-added matrix (Households) and use that row as an economic indicator. If you choose this approach, you do not need to follow the steps for constructing the Q-row, as your economic indicator will already be structured in a way that matches the sectors in your IO table.

Make sure that you include labels and units for all 3 indicators in your satellite block.

EXPLANATION: How can you include data on different indicators in one satellite block (100 words)? Let's imagine there are two satellite accounts with data on greenhouse gas emissions and employment (one satellite account for Country A and another one for Country B). Would the satellite account data for these two countries be the same? Why? Why not? (100 words)

PHYS5033 organisation – assessment tasks

Tips for the Homework tasks:

- Make sure that you read the homework instructions thoroughly for each week – if you don't complete one of the steps expected, you won't get any marks for it.
- If there is a question posed in one of the homework steps, make sure you provide an explicit answer for that question in the Excel file.
- Always include labels and units whenever you present results and make sure that your workings are included.
- You will be provided with worked examples and tutorial answers. These provide you with examples of how the calculations work – they do not provide you with a template for the homework. If you just replicate these, you will not be answering the steps expected in the homework.
- You should be able to answer the questions posed in the 'Explanation' part of the homework each week using the content provided in the lectures and in the background reading material. There is no need to use other sources to answer these questions, and doing so often results in students providing irrelevant or incorrect answers. If you do use other sources of information, these sources must be cited and referenced appropriately. You are strongly advised NOT to rely on AI tools to source your answers for these questions since they usually provide incorrect information.
- Make sure that your Excel submission is easy for the marker to read and find your results. This includes using appropriate font sizes, using appropriate row heights and column widths, and including text to note which step the results you're presenting relate to.

Key takeaways

- A satellite account, or **Q** matrix, can be added to an input-output table in order to calculate a footprint
- Raw data for use in a **Q** matrix come from a variety of sources
- The raw data for a satellite account may not be in the same sector structure as the input-output table
- Data can be converted to the appropriate structure using a concordance, a proxy, and mapping

More to come:

- Calculating footprints using the **Q** matrix